DATA ITEM DESCRIPTION

Title: STRUCTURAL COUPLING TEST REPORT

Number: DI-SESS-82492
Approval Date: 20250709
AMSC Number: F10578
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZFS)
Project Number: SESS-2025-013
Applicable Forms: N/A

Use/Relationship: The Structural Coupling Test Report provides the data for the detailed evaluation of the sensitivity of the air vehicle's automatic flight control system to structural response and follow-on aeroservoelastic analyses. The principal use is to validate or update analytical models.

a. This Data Item Description (DID) is applicable to any contract for which an automated flight control system is being developed, the flight control laws are being modified for said system, or a structural coupling test (SCT) is required.

b. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in section 5 of MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)* (copies of this document are available online at https://quicksearch.dla.mil).

c. This DID supersedes DI-MISC-81387, *Airframe Rigidity Test Report* (copies of this document are available online at https://quicksearch.dla.mil).

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Structural Coupling Test Report shall be in the contractor's format.

3. Content. The Structural Coupling Test Report shall describe the contractor's analyses and test activities associated with characterizing the flight control systems open and closed loop response to the elastic modes of the air vehicle structure. The report shall contain the results and findings from the analyses and tests used to validate and update the aeroservoelastic analyses.

a. The Structural Coupling Test Report shall be based on the interpretation and evaluation of tests and analyses and shall include the following:

(1) Descriptions and photographs of the suspension system, test instrumentation, special test equipment, and air vehicle configurations, including fuel states, landing gear configurations, and store loadings. Differences in the mass and structural configuration of the flight test vehicle and the SCT air vehicle shall be noted.

(2) Descriptions of the methods used for selecting test instrumentation types, test instrumentation capabilities, and test measurement locations.

(3) Description of the methods used for orienting sensors to airframe's coordinate system (e.g., mounting blocks versus coordinate transformation).

(4) Description of the methods used for mounting instrumentation.

(5) Specifications for all test instrumentation.

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(6) Descriptions of the test procedures, test conditions, test methods, data acquisition system, and methods of data reduction analysis.

(7) Copy of the final version of the test plan.

(8) Description of the version of the flight control laws used, including any test-specific modifications.

(9) Description of the automatic flight control system architecture for all axes, including the location of sensors, as well as diagrams and schematics of software filters, test signal injection points, and loop breaks.

(10) Description of the methods used for selecting test cases, including nominal and failure states.

(11) Copy of the test matrix as planned to be executed.

(12) Definition of the measurement units including axis labels for plots and figures.

(13) Plots comparing measured control law frequency response in all axes (bode plots) to pre-test analysis.

(14) Plots of the structural transfer functions versus frequency for accelerations and rates.

(15) Copy of the test log as executed.

(16) Comparison of the measured gain and phase margin results for testing with the pre-test analysis, including the resolution of all test discrepancies and anomalies.

End of DI-SESS-82492

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